

Yu-Chun Lin (June)

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Education

University of Pittsburgh

Professional MS Bioengineering – Neural Engineering Track

Expected graduation: Dec 15, 2026

Aug 2025 – present
Pittsburgh, Pennsylvania

Hong Kong University of Science and Technology (HKUST)

Bachelor of Engineering in Computer Engineering

Sep 2020 – Jun 2024
Hong Kong

Experience

Prof. Nyein's Biosensor Lab at HKUST

Research Assistant

July 2024 – Jun 2025
Hong Kong

- Developed algorithm to extract ECG-Derived Respiratory (EDR) data with correlation coefficient of 0.72.
- Developed a user-friendly interface for a novel sleep-monitoring patch, enabling clinical trial data processing and improving usability.

Sengital Limited – AI & IT Team

AI Research Engineer (6-month Co-op Internship)

Summer 2024
Hong Kong

- Devised novel time series forecasting model and improved the R-squared score from 55.1% to 85.7%.
- Designed and standardized shared-bike user data processing protocol to enhance data features in model training.

Projects

EEG EmotionNET – CNN-SNN hybrid model for emotion recognition

- Developed a bio-inspired architecture using PyTorch and snnTorch for cross-subject EEG emotion recognition, achieving 69.5% classification accuracy on blinded test data.
- Conducted EEG preprocessing, temporal-spatial feature extraction, and signal analysis on the DREAMER and SEED-IV datasets using MATLAB and Python.

Spring 2026
United States

Automatic Pet Feeder - Embedded system project built with STM32-Microcontroller

- Integrated GPS sensor and LoRa module to track and transmit the pet's location.
- Implemented communication protocols such as USART and I2C for seamless interaction with external sensors.

Summer 2023
Taiwan

Publications

Li, Y., Wang, X., Lin, Y., Veronica, A., & Nyein, H. Y. Y. (2025). Multimodal Sponge-Based Wearable Sensor for Continuous Monitoring of Electrochemical and Electrophysiological Signals During Exercise. *Journal of Materials Chemistry B*.
<https://doi.org/10.1039/D5TB01538C>

Sep 2025

Li, Y., Lin, Y., He, A., Wang, X., Wang, X., Nyein, H., & Cui, J. (2026). Intermetallic-anchored epidermal EGaIn patch with analog constriction gates for cardiorespiratory monitoring. *Science Advances*.

Under review

Technical Skills and Academic Interests

Programming Languages: Python, , MATLAB, C++/C, Javascript, HTML, CSS, Reactjs, Nodejs

Courses: Neural Signal Analysis & Processing, Intro. to Neuro-engineering, Medical Device Ideation & Development, Human Factor Engineering, Stochastic Analysis.

Dev Tools: VScode, Git, Github, JupyterLab, CubeIDE, Solidwork, Prism GraphPad.

Areas of Interest: Brain-machine interface; Real-time decoding; Unsupervised machine learning.

Languages: Mandarin (Native), English (Professional), Cantonese (fluent), Korean (Elementary)